RAVEN-Select: Value and Limits of OCR-Grounded Answer Selection for Budgeted Document VQA

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Abstract—High-resolution document visual question answering (Document VQA) is commonly approximated under limited visual budgets by either resizing the full page or selecting a few question-relevant patches. However, better evidence retrieval does not necessarily yield a better final answer, and no single reader is uniformly reliable. We propose RAVEN-Select, a reader-aware, OCR-grounded answer-output selector. Three budgeted readers generate answers from a resized page, BM25-selected patches, and learned-ranker-selected patches. RAVEN-Select discards empty outputs, identifies generated answers whose normalized strings occur in page or selected-patch OCR, and returns the shortest grounded answer; if none is grounded, it falls back to the shortest nonempty output. The selector uses no gold-derived feature at inference. On a 1,000-question DocVQA validation subset, RAVEN-Select obtains 0.8234 ANLS and 0.723 EM, compared with 0.8149/0.706 for resize and 0.8173/0.708 for the strongest simple equal-cost heuristic. The paired ANLS gain over shortest nonempty is significant (95% CI $[+0.0008, +0.0123]$), while the interval against resize crosses zero ($[-0.0019, +0.0199]$); the prespecified scale gate is therefore partial and full-validation evaluation is not run. On a second VLM, the method again significantly improves over resize but not over shortest nonempty. Changing grounding from EasyOCR to Tesseract leaves all decisions unchanged on $n = 500$. Frozen-rule transfer gates fail on InfographicVQA and an MP-DocVQA contact-sheet adaptation; on InfographicVQA, BM25 alone is substantially stronger than the selector. These results show both the value and the limits of output grounding and selection in budgeted Document VQA.

Index Terms—Document visual question answering, answer selection, OCR grounding, visual language models, budgeted inference, model ensembles.

I. Introduction

DOCUMENT visual question answering requires reading text from high-resolution, layout-rich images and returning a short answer to a natural language question. It builds on text-rich visual question answering, where models must not only perceive visual content but also read and copy short textual answers from image text [1]–[3]. Modern visual language models (VLMs) can process such documents directly, but their visual-token and memory costs grow with image resolution. Practical systems therefore resize an entire page, select a small number of patches, or compress the visual input [4]–[9]. This creates a transfer problem: a retrieval method may improve evidence coverage without improving the answer generated by the downstream reader.

Our experiments expose this gap. A learned evidence ranker improves patch coverage, yet its end-to-end answer quality trails a simple resized-page reader. A retrieve–extract–verify diagnostic achieves high candidate recall but weak top-1 ranking. Multiple-choice verification is worse than free generation. Finally, a learned post-hoc route regressor improves results on 300 examples but does not significantly beat resize at 500 examples and loses to a simple shortest-nonempty equal-cost ensemble. These findings shift the central question from which patches should be selected? to which generated answer should be trusted?

We introduce RAVEN-Select (Reader-Aware, OCR-Grounded Answer Selection), a deterministic output selector for a small set of budgeted readers. The readers observe complementary views: a resized full page, question-selected lexical patches, and learned-ranker-selected patches. RAVEN-Select validates each generated output against OCR text already available from the page or selected patches, then uses answer conciseness to resolve multiple grounded candidates. The method is deliberately simple. Its novelty lies in treating OCR as an inference-time grounding signal for generated answers, not as a gold-answer diagnostic or as another retrieval target. The contribution is not a new VLM architecture; it is a controlled study showing that OCR-grounded output selection can improve over equal-cost answer-selection heuristics on DocVQA, while its advantage over strong one-call resize readers and its cross-dataset transfer are limited. The central claim is that, under a fixed set of budgeted readers, selecting the grounded final answer can be more effective than further improving patch retrieval, but the strength of this effect is dataset-dependent.

Our contributions are:

- We formulate budgeted Document VQA as answer-output selection among complementary readers rather than as patch selection alone.
- We propose a leakage-safe OCR-grounded selector that combines generated answer presence and conciseness without using gold answers, oracle routes, or gold-derived OCR flags.
- We evaluate against single readers, equal-cost heuristics, the previous learned route regressor, and four learned selectors using question-level out-of-fold evaluation.

- We scale from n=500 to n=1000 and report a partial result: RAVEN-Select significantly improves ANLS over the strongest equal-cost heuristic and improves over resize in mean, but does not significantly beat resize; we therefore do not proceed to full validation.
- We report exact component, route, cost, second-VLM, second-OCR, and cross-dataset analyses, including negative results rather than presenting a learned selector as the main contribution.

II. Related Work

A. Document VQA and Document Reading

DocVQA established question answering over scanned and born-digital document images [1]. TextVQA and M4C showed that reading visible text and copying short OCR tokens are central challenges in visual QA [2], [3]. A separate line of document AI work learns stronger representations by combining OCR text, layout, and vision, as in LayoutLM and its successors [10]–[13], or removes the external OCR dependency with image-to-text readers such as Donut and Pix2Struct [14], [15]. RAVEN-Select differs from both: it freezes the readers and improves the answer-selection interface around their outputs.

B. High-Resolution and Budgeted VLMs

Text-rich images require more spatial detail than ordinary images. Recent systems address this cost through dynamic resolution, slicing, shape-adaptive cropping, document-token compression, structured OCR/layout encoding, or efficient visual encoders [4]–[9]. RAVEN-Select is complementary to these architectural approaches. It neither changes the VLM nor claims a cheaper encoder; it runs a small fixed set of budgeted readers and selects their final answer.

C. Budgeted Visual Evidence Selection

When a full-resolution page exceeds the visual budget, systems may retrieve text or image regions. BM25 remains a strong lexical retrieval baseline [16], while document-image retrieval methods such as ColPali demonstrate the value of late-interaction visual representations [17]. Retrieval-augmented VQA studies the interaction between evidence retrieval and generation [18], and reader-centered passage selection argues that query relevance and reader usefulness are not identical [19]. Our BM25 and learned evidence routes each expose an overview plus $K = 2$ selected patches to the VLM. Unlike retrieval work, our method does not claim improved evidence ranking; it selects among answers produced by fixed readers.

D. Answer Reranking and Ensembles

Answer reranking has been studied in textual QA, including learned candidate ranking in RankQA [20]. LLM-Blender uses pairwise ranking and generative fusion to exploit complementary model outputs [21], while self-consistency selects an answer supported across multiple generated reasoning paths [22]. Cost-aware methods such as RouteLLM and FrugalGPT route queries or build cascades to trade cost against quality [23], [24]. RAVEN-Select is closest in spirit to output selection, but its routing unit is a document-reader answer and its decision signal is OCR grounding plus conciseness. We compare learned regression and LambdaRank selectors (implemented with LightGBM [25]) against transparent equal-cost rules. The central comparison is not against a weak majority vote but against the strongest observed simple heuristic, shortest nonempty.

III. Problem Formulation

Let q be a question, I a document image, and

$$\mathcal{R} = \{r_{\text{resize}}, r_{\text{BM25}}, r_{\text{LER}}\}$$

the fixed reader set. Each reader returns a string $y_r = f_r(I, q)$ under its visual input policy. The output selector $s(q, I, \{y_r\})$ chooses one existing answer; it does not synthesize a new answer. Let C_r denote reader cost. The post-generation setting has $C_{\text{select}} = \sum_{r \in \mathcal{R}} C_r$ plus OCR and negligible string selection. Our objective is higher expected ANLS and exact match (EM) than single readers and selectors with the same three-reader cost.

The setting differs from pre-routing. A pre-router chooses one reader before generation and costs one VLM call. RAVEN-Select is accuracy-oriented and uses three calls; it is not claimed to be cheaper than resize.

IV. RAVEN-Select

A. Budgeted Reader Set

a) Resize.: The complete page is resized to the VLM input limit and processed in one call. It preserves global context but can make small text difficult to read.

b) BM25 patches.: OCR text from page regions is scored against the question using BM25 [16]. The VLM receives a page overview and the top $K = 2$ patches.

c) LER-BOPS patches.: A learned evidence ranker selects $K = 2$ patches from the same budgeted pool. It was developed to improve evidence coverage, but it is retained here as a complementary reader rather than claimed as the best standalone answerer.

B. Normalization and Grounding

For a string y, the normalization function $N(y)$ lowercases text, removes punctuation, strips leading and trailing whitespace, and collapses repeated whitespace. The same normalization is applied to page OCR T_{page} and route-specific selected-patch OCR $T_{\text{patch},r}$.

An output is grounded when its nonempty normalized form is a substring of either OCR source:

$$g_r = \mathbb{1}\begin{bmatrix} N(y_r) \neq \emptyset \; \wedge \\ \big(N(y_r) \subseteq N(T_{\text{page}}) \\ \quad \vee \; N(y_r) \subseteq N(T_{\text{patch},r})\big) \end{bmatrix}. \tag{1}$$

Here $\subseteq$ denotes contiguous string occurrence, not token-set inclusion. This conservative check supports exact textual

Algorithm 1: RAVEN-Select
Input: question q, page I, outputs $\{y_{\text{resize}}, y_{\text{BM25}}, y_{\text{LER}}\}$, page OCR, selected-patch OCR.
1) Normalize all outputs and OCR text.
2) Discard empty outputs.
3) Mark an output as OCR-grounded if it occurs in page or route-patch OCR.
4) If grounded outputs exist, return the shortest grounded output.
5) Otherwise, return the shortest nonempty output.
6) Break ties by resize, then BM25, then LER-BOPS.

Fig. 1. RAVEN-Select is a deterministic inference-time selector.

answers such as dates, amounts, names, and short phrases. It intentionally does not use semantic similarity, which could introduce another learned component and obscure the source of gain. Importantly, presence is checked against OCR extracted from the input document, not against an "answer OCR" field or any text derived from the gold answer. For example, if one reader outputs "Total Amount Due: 48.50" and another outputs "48.50," both may be grounded in the document OCR, but the shorter grounded output is preferred.

C. Selection Rule

RAVEN-Select follows answer reranking and multi-output selection rather than patch selection alone [20]–[22]. Unlike a learned pairwise ranker or agreement vote, it uses a document-specific grounding predicate and returns one existing reader output. Let $\ell_r = |N(y_r)|$ and let $\mathcal{G} = \{r : g_r = 1\}$. RAVEN-Select returns

$$r^\star = \begin{cases} \arg\min_{r\in\mathcal{G}} \ell_r, & \mathcal{G} \neq \emptyset, \\ \arg\min_{r:N(y_r)\neq\emptyset} \ell_r, & \text{otherwise.} \end{cases} \tag{2}$$

Exact length ties are broken deterministically in the order resize, BM25, then LER-BOPS. If all outputs are empty, resize is returned. The first branch rewards answers supported by OCR; the minimum-length rule penalizes verbose VLM continuations and surrounding sentence text. The second branch is the strongest simple equal-cost baseline and supplies a stable fallback when OCR is incomplete.

D. Leakage Safeguards

The grounding check compares the generated prediction with OCR extracted from the input image or selected patches. Gold answers are used only after selection to compute ANLS/EM. We explicitly exclude:

- gold answer text or embeddings;
- ANLS or exact match to gold as inference features;
- answer_in_full_image_ocr and answer_in_selected_patch_ocr diagnostic columns, because those flags were computed using gold answers;
- oracle route identities or any feature derived from ground-truth correctness.

The OCR-presence cache stores the prediction, OCR text, and prediction-presence flags per (question, route, VLM). Model tags are part of the cache key to prevent robustness-model outputs from sharing flags with the primary VLM.

E. Learned Selector Comparators

For completeness, we construct one row per question–route output with leakage-safe route, prediction-text, question-type, OCR-presence, consensus, substring, edit-similarity, and retrieval-confidence features. Training labels are per-output ANLS to gold, but labels are visible only in training folds. Five-fold out-of-fold (OOF) evaluation groups rows by question/image so that all three outputs for a question remain in the same fold. We compare ridge regression, logistic best-route classification, LightGBM regression, and LightGBM LambdaRank. These are comparators and ablations; none is the primary method.

V. Experimental Setup

A. Data and Models

We evaluate nested deterministic subsets of the DocVQA validation split [1]: development sets of $n = 300$ and $n = 500$, and the primary scale gate set of $n = 1000$. The principal VLM is Qwen2.5-VL-3B-Instruct in 4-bit mode [4]. A second-model robustness experiment uses Qwen2-VL-2B-Instruct on $n = 500$. Transfer experiments use InfographicVQA $n = 300$ and an MP-DocVQA contact-sheet adaptation $n = 300$; the latter is not the standard multi-page task. The primary OCR engine is EasyOCR, with Tesseract as a grounding robustness check. Patch readers use a page overview and $K = 2$ selected patches.

These subsets are not presented as a full-dataset state-of-the-art comparison. They are controlled, paired experiments over identical questions and cached reader outputs.

B. Metrics and Statistics

We report Average Normalized Levenshtein Similarity (ANLS) and exact match (EM). All comparisons are paired at question level. Confidence intervals are percentile bootstrap 95% intervals over 1,000 resamples with seed 42. The prespecified $n = 1000$ scale gate requires:

1) ANLS greater than resize and shortest nonempty;
2) CI lower bound greater than zero against both resize and shortest nonempty.

Passing both comparisons permits full-validation evaluation; significance against exactly one baseline is reported as partial and blocks that evaluation.

TABLE I
Prespecified gate outcomes. Significance is on paired ANLS 95% CIs.

Experiment	Setting	Outcome
DocVQA development	$n = 500$	PASS
DocVQA scale	$n = 1000$	PARTIAL†
Second VLM	Qwen2-VL-2B, $n = 500$	PARTIAL‡
Second OCR	Tesseract, $n = 500$	PASS
Transfer	InfographicVQA, $n = 300$	FAIL
Transfer	MP contact sheet, $n = 300$	FAIL

†significant vs. shortest, not resize; ‡significant vs. resize, not shortest.

TABLE II
Primary DocVQA scale result ($n = 1000$). Three-output selectors consume the same cached VLM outputs.

Method	ANLS	EM
Resize (1 call)	0.8149	0.706
BM25 (1 call)	0.7796	0.674
LER-BOPS (1 call)	0.7873	0.681
Shortest nonempty	0.8173	0.708
RAVEN-Select	0.8234	0.723

C. Baselines

Single-reader baselines are resize, BM25, and LER-BOPS. Equal-cost three-reader selectors include normalized majority, resize-on-disagreement, BM25-on-disagreement, shortest nonempty, answer-type shortest, consensus-first, and the previous learned RAVEN-post route regressor. The strongest simple baseline at the primary scale is shortest nonempty. We therefore compare RAVEN-Select only against equal-cost selectors that consume the same three cached reader outputs, and against one-call readers separately.

VI. Results

A. Outcome Summary

Table I consolidates every prespecified gate so the study's scope is visible at a glance. The method fully passed only the original $n = 500$ development gate; at scale, on a second VLM, and across datasets the outcomes are partial or negative. We present these together to make the contribution honestly bounded rather than to imply a broadly reliable method.

B. Main Result

Table II reports the primary scale result. RAVEN-Select improves ANLS by 0.0086 over resize and by 0.0061 over shortest nonempty. All significance tests are on ANLS; EM is reported descriptively, improving from 0.706 (resize) and 0.708 (shortest nonempty) to 0.723 without paired intervals. The paired ANLS interval against shortest nonempty has a positive lower bound, but the interval against resize crosses zero. The prespecified gate is therefore partial, and we do not run or claim a full-validation result.

TABLE III
Paired ANLS differences for RAVEN-Select on $n = 1000$.

Comparison	Mean Δ	95% CI
vs. resize	+0.0086	$[-0.0019, +0.0199]$
vs. shortest nonempty	+0.0061	$[+0.0008, +0.0123]$

TABLE IV
Learned OOF selectors on DocVQA $n = 500$.

Selector	ANLS	EM
Logistic best-route	0.7843	0.674
Ridge regression	0.7992	0.700
LightGBM regression	0.8005	0.698
LightGBM LambdaRank	0.8025	0.690
RAVEN-Select rule	0.8053	0.706

C. Development-Scale Result

On the earlier $n = 500$ development gate, RAVEN-Select obtained 0.8053 ANLS and 0.706 EM versus 0.7840/0.676 for resize and 0.7965/0.688 for shortest nonempty. Both paired ANLS intervals were positive. The weaker scale result above is reported without retuning the frozen rule.

D. Learned Selectors

Table IV compares the learned output selectors on $n = 500$. LambdaRank is the strongest learned mean at 0.8025, but its CI lower bound is negative versus shortest nonempty. The transparent OCR-grounded rule is both more accurate on this development subset. Although learned selectors approach the rule, their gains over shortest nonempty are not statistically stable, and they require fold-trained labels. RAVEN-Select is deterministic, transparent, and does not require selector-label training.

E. Component and Route Ablations

Table V isolates grounding, conciseness, OCR source, and route contributions. Removing OCR reduces the method exactly to shortest nonempty (0.7965). Removing shortest selection while retaining grounding yields 0.8015, showing that both components contribute. Page-only equals the union on this sample; patch-only has slightly higher ANLS but lower EM, while the frozen page-or-patch rule gives the strongest balanced endpoint. Removing either patch reader lowers performance, with LER removal causing the larger drop.

F. Cost and Operating Point

RAVEN-Select is an accuracy-oriented post-generation selector. Table VI reports one call per reader and three calls for the method. The approximate RAVEN-Select runtime is the sum of path medians; it excludes one-time OCR cache construction and includes negligible cached selection time. An uncached deployment incurs additional OCR latency. Accordingly, accuracy comparisons

TABLE V
Exact RAVEN-Select ablations on DocVQA $n = 500$. "No shortest" uses deterministic route priority among grounded outputs.

Selector	ANLS	EM	Interpretation
Resize only	0.7840	0.676	Best single reader
BM25 only	0.7555	0.658	Lexical patch reader
LER-BOPS only	0.7543	0.654	Learned patch reader
Shortest only; no OCR	0.7965	0.688	Removes grounding
OCR-grounded; no shortest	0.8015	0.702	Removes conciseness
Page-OCR + shortest	0.8053	0.706	Full-page grounding only
Patch-OCR + shortest	0.8065	0.698	Patch grounding only
Page-or-patch OCR + shortest	0.8053	0.706	Frozen RAVEN-Select
No BM25 route	0.8023	0.700	Resize + LER only
No LER route	0.7962	0.696	Resize + BM25 only
Best learned selector	0.8025	0.690	LambdaRank, 5-fold OOF

TABLE VI
Honest cost accounting on $n = 1000$.

Method	VLM calls	Median s	ANLS	EM
Resize	1	2.609	0.8149	0.706
BM25	1	3.141	0.7796	0.674
LER-BOPS	1	5.178	0.7873	0.681
Shortest nonempty	3	~ 10.928	0.8173	0.708
RAVEN-Select	3	$\sim$ **10.928**	0.8234	0.723

TABLE VII
Second-VLM robustness ($n = 500$, Qwen2-VL-2B).

Method	ANLS	EM
Resize	0.5913	0.526
BM25	0.4787	0.406
LER-BOPS	0.4937	0.418
Shortest nonempty	0.6124	0.540
RAVEN-Select	0.6128	0.544

distinguish equal-cost three-output selectors from one-call reader operating points. Comparisons among shortest nonempty, learned selectors, and RAVEN-Select are equal-cost three-call settings; resize, BM25, and LER-BOPS remain the one-call operating points.

The one-call pre-answer router is retained as a negative result: on $n = 500$ it reaches 0.7762 ANLS versus 0.7840 for resize. This supports the hypothesis that post-generation output information is important, but it also means RAVEN-Select should not be described as a budget-saving pre-router.

G. Second-VLM Robustness

Table VII evaluates the same selector with Qwen2-VL-2B-Instruct on $n = 500$. RAVEN-Select improves over resize by 0.0215 ANLS with CI $[+0.0049, +0.0393]$. It is essentially tied with shortest nonempty: the mean gain is 0.0005 and the lower confidence bound is zero. We therefore treat this as partial robustness: it preserves the improvement over the single resize reader, while showing that the incremental OCR-grounding advantage over shortest-answer selection is model-dependent.

H. OCR-Engine Robustness

Holding all DocVQA $n = 500$ reader outputs fixed, replacing EasyOCR with Tesseract for the grounding predicate leaves the RAVEN-Select route choice unchanged for every example. Both engines therefore obtain 0.8053 ANLS and 0.706 EM (drop 0.0000), passing the pre-specified maximum-drop threshold of 0.015. This checks the grounding interface, not end-to-end OCR-conditioned reader generation.

I. Dataset Transfer

Table VIII applies method version 1.0.0 without re-tuning. RAVEN-Select has positive mean gains over re-size and shortest nonempty on both datasets, but all relevant intervals cross zero, so both transfer gates fail. On InfographicVQA, BM25 alone is substantially stronger than the selector, showing that the frozen resize-favoring decision rule does not transfer to this reader ordering. The MP-DocVQA experiment is only a contact-sheet adaptation: pages are rasterized into one image before applying the same budgeted pipeline.

For InfographicVQA, RAVEN-Select's ANLS intervals are $[-0.0125, +0.0377]$ against resize and $[-0.0049, +0.0275]$ against shortest. For the MP contact sheet, the interval against resize crosses zero ($[-0.0095, +0.0330]$), while the gain over shortest is significant ($[+0.0024, +0.0238]$). The preregistered partial-transfer condition requires significance over resize; shortest-only significance therefore remains FAIL.

VII. Analysis

A. Oracle Ceiling and Headroom

Because RAVEN-Select can only choose among three outputs, its performance is bounded by the best-of-three reader oracle. Table IX reports the oracle ceiling. The available headroom over resize is 0.0398 ANLS, while RAVEN-Select recovers 0.0086 ANLS, or 21.5% of the recoverable gap.

B. Normalization Ablations

Table X compares grounding normalizations on identical cached outputs. The conservative rule remains the frozen

TABLE VIII
Frozen-rule dataset transfer ($n = 300$ each). MP denotes the contact-sheet adaptation, not standard multi-page MP-DocVQA.

Dataset	Resize	BM25	LER-BOPS	Shortest	RAVEN-Select	Gate
InfographicVQA	0.3047	0.3713	0.3511	0.3053	0.3167	FAIL
MP contact sheet	0.3252	0.3065	0.3288	0.3244	0.3360	FAIL

TABLE IX
Oracle ceiling and headroom on DocVQA $n = 1000$.

Method	ANLS	EM
Resize	0.8149	0.706
RAVEN-Select	0.8234	0.723
Best-of-3 oracle	0.8547	0.759

TABLE X
Grounding normalization ablations ($n = 1000$).

Variant	ANLS	EM
RAVEN-Select-Raw	0.8187	0.715
RAVEN-Select (conservative)	0.8234	0.723
RAVEN-Select-NumNorm	0.8245	0.720
RAVEN-Select-DateNorm	0.8094	0.703
RAVEN-Select-Fuzzy	0.8240	0.716

production method; all other rows are exploratory variants and retain their distinct names.

C. Answer-Type Breakdown

Table XI summarizes performance by answer category. Gains are most visible for numeric and person/name questions, while address/location is unchanged across selectors.

D. Failure Taxonomy and Qualitative Examples

We applied a frozen heuristic taxonomy over 100 stratified questions (label_source=heuristic_v1). Table XII lists the primary-label counts. Secondary labels appear for S1 (7), S2 (21), S6 (2), and F8 (7), reflecting ancillary factors rather than the main cause. Labels are a frozen heuristic pass and should not be interpreted as human-adjudicated error annotations.

Qualitative checks align with these buckets. BM25 correctly selects “Fred L. Smith, Jr,” while LER-BOPS supplies “Dr. Arley T. Bever” and “Definitions” in three S3 examples. Failures include selecting an unrelated organization for an addressee question (F1), an OCR-grounding miss even when “San Diego” is generated (F2), and choosing “Severe” instead of “Very Mild” (F3). These heuristic cases require human audit, but illustrate why grounding and conciseness cannot exceed the best-of-reader ceiling.

VIII. Discussion

A. Why Grounding and Conciseness Help

Document answers are frequently short strings copied from the page: names, dates, totals, percentages, or addresses. VLM readers can identify the right region yet produce extra words, explanations, or nearby values. OCR presence rejects unsupported generations, while shortest selection removes surrounding language. The combination targets final-answer reliability rather than retrieval coverage.

TABLE XI
Answer-type breakdown on DocVQA $n = 1000$.

Answer type	Support	Resize	Shortest	RAVEN-Select
address/location	23	0.8203	0.8203	0.8203
amount/currency	62	0.7776	0.7833	0.7833
date	126	0.8560	0.8660	0.8691
id/code	17	0.8497	0.8864	0.8864
number	75	0.8233	0.7963	0.8363
organization	25	0.7933	0.8379	0.8406
person/name	184	0.7976	0.8002	0.8137
phrase/other	488	0.8138	0.8151	0.8155

TABLE XII
Heuristic taxonomy counts (primary labels, $n = 100$).

Label	Meaning	Count
S3	Correct patch answer	29
S4	Correct resize answer	2
S6	Reader disagreement resolved	7
F1	No reader correct	21
F2	OCR miss	6
F3	Wrong shorter span	7
F5	Normalization mismatch	3
F6	Non-verbatim paraphrase	13
F7	Incomplete answer	6
F8	Annotation ambiguity	6

B. What the Research Arc Reveals

The sequence of negative and positive findings is informative. Patch retrieval alone improved evidence coverage but not final ANLS. Candidate extraction reached recall@20 near 0.88 while top-1 ranking remained weak. Multiple-choice verification underperformed free generation. Learned post-hoc routing worked on $n = 300$ but weakened at $n = 500$. The successful intervention acts at the last transfer step: selecting a grounded final output. Thus, in this budgeted setting, answer grounding matters more than another patch-retrieval refinement. Our raw OCR-text augmentation was also not the winning interface. This is consistent with work arguing that structured OCR/layout compression can be more effective than simply appending raw OCR strings to a VLM [9].

C. Why Transfer Fails

The failed InfographicVQA transfer suggests that RAVEN-Select is not a universal output selector. Its rule assumes that concise OCR-grounded spans are preferable and that resize remains a reliable global reader. In InfographicVQA, BM25 is the strongest single path (0.3713 ANLS versus 0.3167 for the frozen selector), indicating that dataset-specific reader ordering can dominate the simple selection rule. Because the tie-break and fallback order (resize → BM25 → LER-BOPS) and the shortest-answer preference were frozen on DocVQA, they are over-aligned to DocVQA output patterns and do not adapt when a different reader is dominant. This points to an adaptive-ordering variant as future work, which must be trained and validated on held-out data rather than retuned on these transfer results.

D. Simplicity Versus Learned Routing

RAVEN-Select has no learned parameters. This is a strength only because the comparison is fair: all three-reader selectors consume the same cached outputs, and the simple method beats learned regressors on the $n = 500$ development comparison and significantly beats shortest nonempty at $n = 1000$. The result should not be interpreted as evidence that learned selection is generally unnecessary; rather, the available labels and feature set do not justify its additional complexity here.

IX. Limitations

- Three VLM calls. RAVEN-Select improves accuracy but is more expensive than a one-call resize reader. It is an equal-cost multi-reader selector, not a cheap pre-router.
- Subset scale. On 1,000 DocVQA questions, the gain is significant against shortest nonempty but not resize. Because the prespecified gate is partial, we do not evaluate the full validation split. These subsets are not a state-of-the-art benchmark claim.
- Second-model uncertainty. Qwen2-VL-2B retains the gain over resize but not a positive CI lower bound over shortest nonempty.
- OCR dependence. Exact substring grounding can reject correct paraphrases and inherit OCR errors. Tesseract reproduces the EasyOCR decisions on $n = 500$, but this single robustness check does not establish engine invariance.
- Best-of-reader ceiling. Because RAVEN-Select chooses among generated outputs, it cannot recover an answer that none of the readers generate. Its ceiling is therefore bounded by the best-of-reader oracle, which reaches 0.8547 ANLS on $n = 1000$ versus 0.8234 for the method.
- Language and answer form. The method is best suited to extractive short answers. Long-form, abstractive, multilingual, or heavily normalized answers may not occur verbatim in OCR.
- Patch-only ANLS. Patch-only grounding is 0.0012 higher in mean ANLS on $n = 500$ but 0.008 lower in EM than the frozen union rule. Broader evaluation is needed to determine whether this trade-off generalizes.
- Transfer. Both $n = 300$ transfer gates fail, and BM25 alone outperforms the frozen selector on InfographicVQA. The MP-DocVQA experiment rasterizes pages into a contact sheet; it is not the standard multi-page benchmark and does not show that RAVEN-Select solves MP-DocVQA.

X. Conclusion

We presented RAVEN-Select, a simple OCR-grounded answer-output selector for budgeted Document VQA. Instead of further optimizing patch retrieval, the method runs three complementary readers and selects the shortest nonempty generated answer supported by page or patch OCR. On DocVQA $n = 1000$, it reaches 0.8234 ANLS versus 0.8149 for resize and 0.8173 for shortest nonempty. Only the latter gain is statistically supported, so the scale gate is partial and full validation remains blocked. Second-model and second-OCR checks show limited robustness, while failed InfographicVQA and MP contact-sheet gates caution against broad generalization. When better evidence does not transfer to better answers, grounding and output selection remain useful design targets, not a universal replacement for stronger readers.

Reproducibility

All selectors operate on cached per-question outputs. The main commands are:

```
python scripts/run_raven_select_build_features.py --n 1000
python scripts/run_raven_select_eval.py --n 1000 --write-gates
python scripts/run_raven_select_paper_ablations.py --n 1000
```

Primary artifacts are raven_select_overview_n1000.json, raven_select_paper_ablations_n1000.json, and the P17–P24 gate and analysis reports.

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